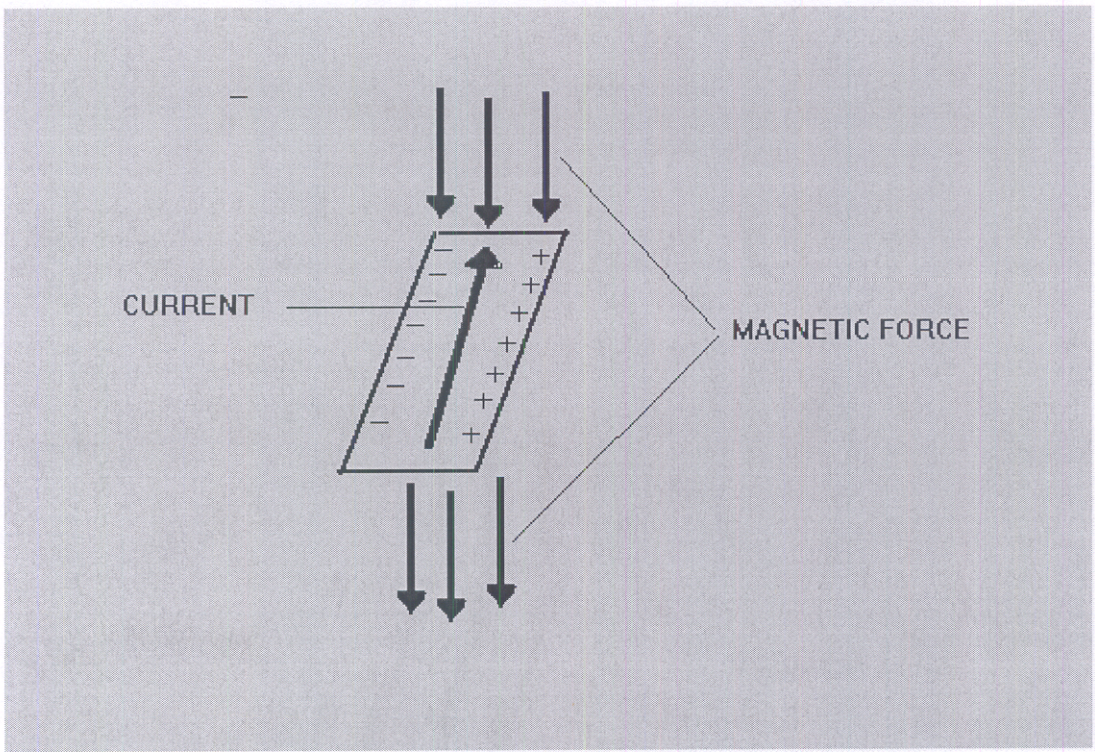
HE HALL EFFE

by Ken Hatton
and
Harold Fennel1

When a current-carrying electrical conductor is placed in a magnetic field, a voltage sometimes develops between one side of the conductor and the other. For this to happen, the magnetic lies of force must be perpendicular, or nearly perpendicular, to the line containing the conductor. The voltage then appears at right angles to the magnetic lines of force. If the conductor is a strip, and the magnetic lines of force are perpendicular to the strip, then the voltage will appear between opposite edges of the strip (See figure one). This is called the Hall effect.



The electric field intensity given as e, is described by the formula:

$$e = kim$$

where i is the current through the conductor, m is the magnetic field strength, and k is the Hall constant.

In some metals the voltage displays a polarity opposite to that of other metals under the same conditions. The polarity depends upon the atomic structure of the metal. The Hall effect occurs in semiconductors as well as metals.

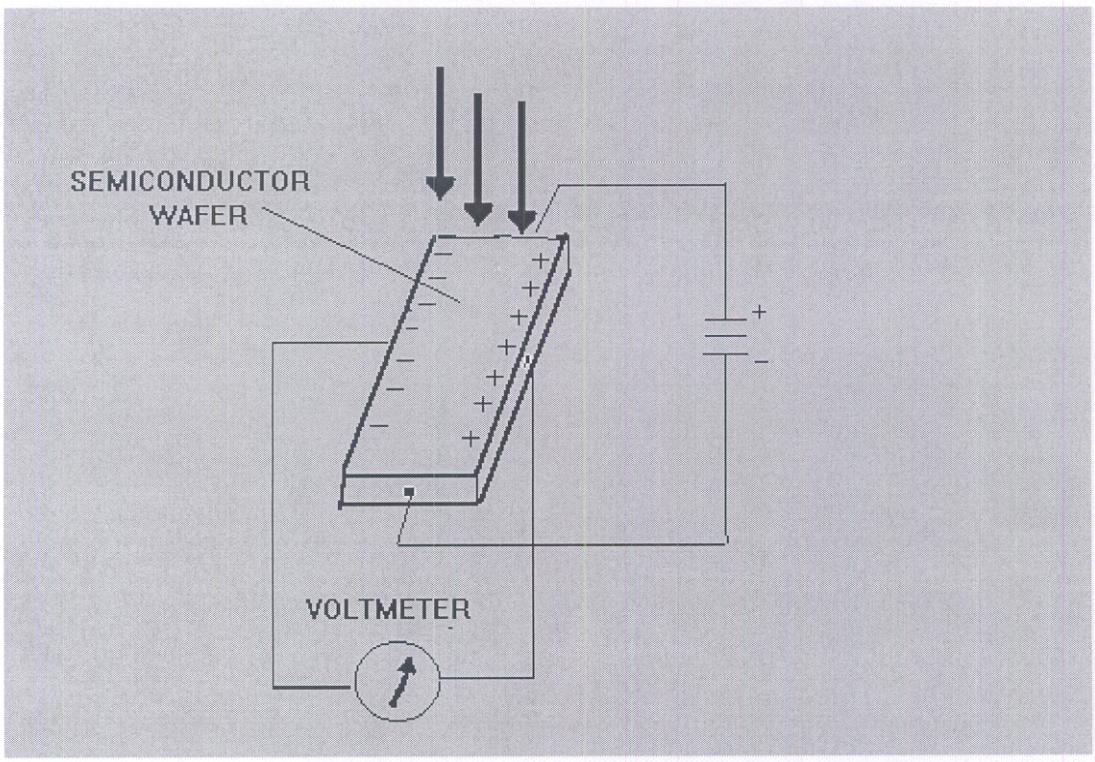
Hall Mobility is an expression of the extent to which the Hall effect takes place in a semiconductor materialFor a given magnetic field intensity and current value, the voltage generated by the Hall effect is greater when the Hall Mobility is higher.

The Hall Mobility is given by the product of the Hall constant and the conductivity for a given material. In general, the greater the carrier

mobility in a semiconductor, the greater the Hall mobility

GENERATOR

A Hall generator is a device that makes use of the Hall effect for the purpose of generating a direct current voltage in the presence of a magnetic field A semiconductor wafer is used as a conductor. A battery, or other direct current source is connected to opposite ends of the wafer. A voltmeter is connected to the adjacent sides of the wafer. The voltage will be proportional to the magnetic field it is exposed to. (see (figure two).

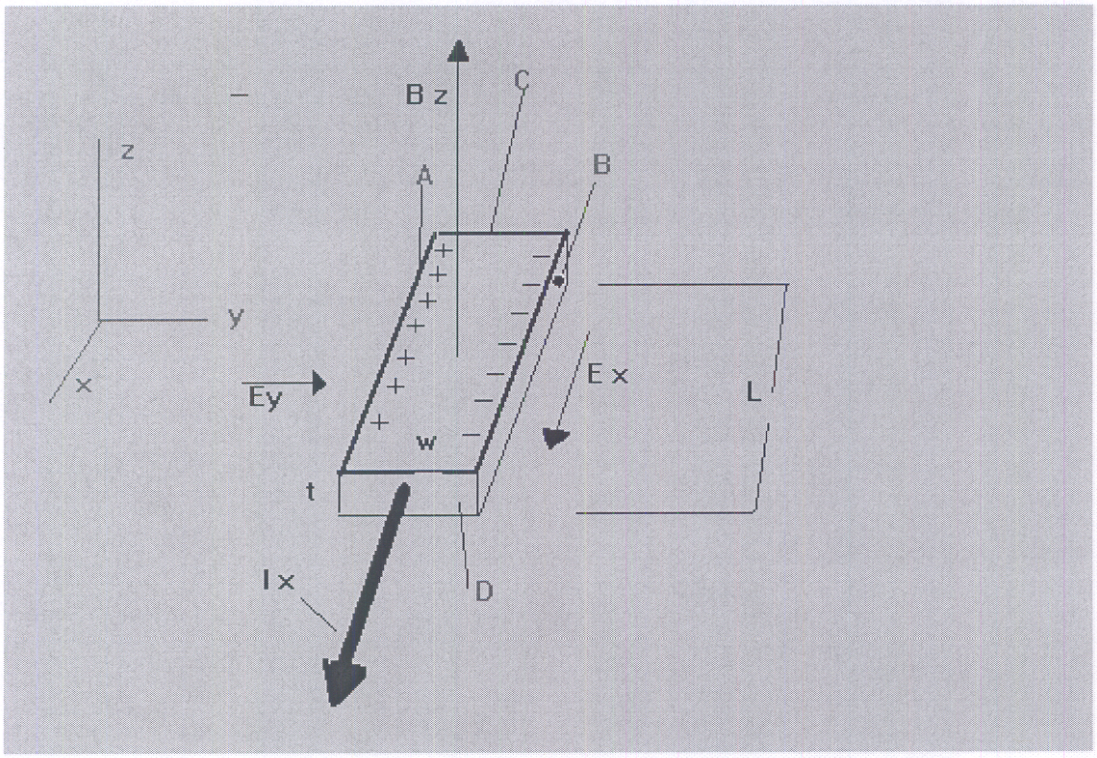


As stated earlier some semiconductor materials are much better than others for use in this application. The better the Hall Mobilitv the better the observation of the Hall effect. Indium antimonide is especially effective for this purpose.

When a magnetic field exists in the vicinity of the wafer, such that the magnetic lines of force are perpendicular to the wafer. a voltage is induced and will register on the voltmeter. The value of the voltage is quite small, but is proportional to the current in the wafer and to the intensity of the magnetic field. This application is an effective means of measuring the intensity of a magnetic field.

A TECHNICAL APPROACH TO THE HALL EFFE

Let's look at what the Hall effect is from a more technical point of view. If a magnetic field is apphed perpendicular to the direction in which holes drift in a P-type bar, the path of the holes tend to be deflected (see figure three). Using vector notation, the total force on a single hole due to the electric and magnetic field is given by:



(1) $F=q(E + V \times B)$

In the y direction the force is:

(2) $Fy = q(Ey - VxBx)$

The basic interpretation of equation 2 is that unless an electric field Ey is established along the width of the bar, each hole will experience a net force(and therefore an acceleration) in the -y direction due to the qVxBz product. Therefore to maintain a steady state flow of holes down the length of the bar, the electric field Ey, must just balance the product VxBz. Essentially:

$$Ey=VxBz$$

so that the net force Fy is zero. Physically this electric field is set up when the magnetic field shifts the hole distribution slightly in the -y direction. Once the electric field Ey becomes as large as VxBz, no net lateral force is experienced by the holes as they drift along the bar. The establishment of the electric field Ey is known as the Hall effect, and the resulting voltage VAB= Eyw is known as the Hall voltage. The electric field Ey can be represented by

$$Ey = RHJxBz$$

The current density J that results from the net drift or movement of holes is just the number of holes crossing a unit area per unit time. B is the magnetic field, and RH is the proportionality constant known as the Hall constant or Hall coefficient.

RH is defined by:

$$RH = 1/qpo$$

po the concentration of holes in the valence band. q is the + charge. Another way of thinking of po is remembering that in n type semiconductors electrons are the charge carriers and in p type holes are charge carriers. Think of a hole as an empty state in the valence band.

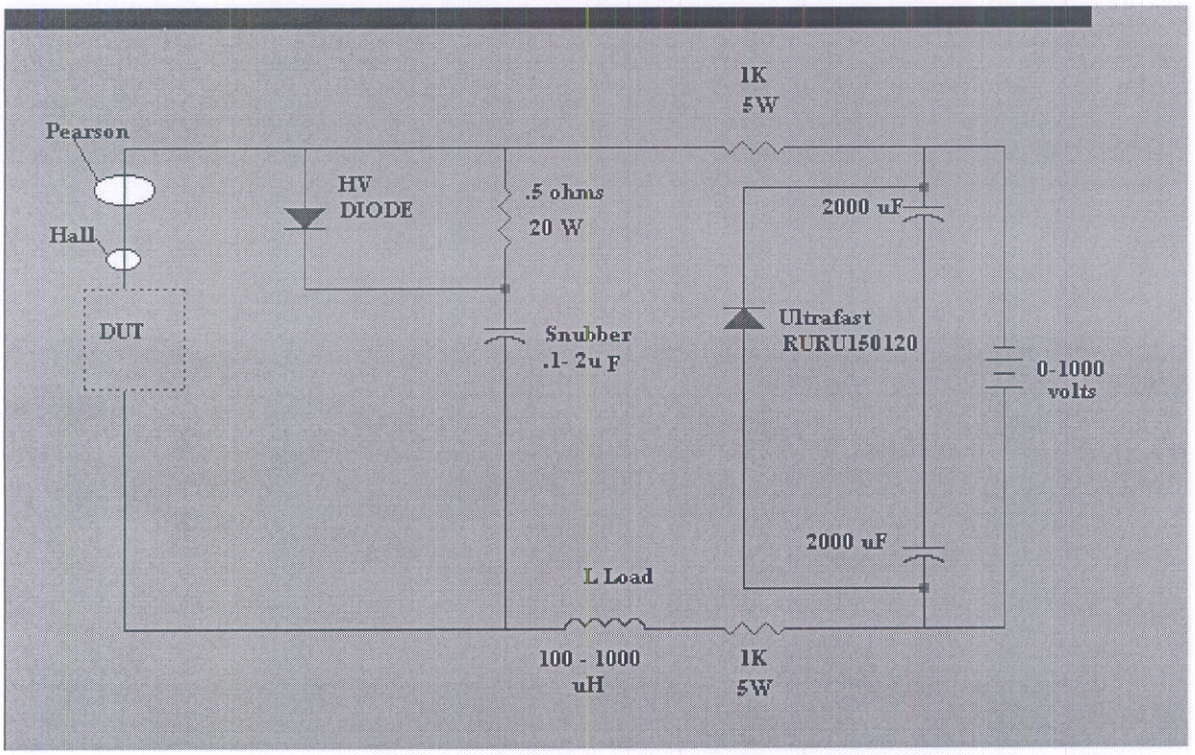
Although the discussion here has been related to p type material, similar results are obtained for n type. A negative value of q is used for electrons, and VAB and RH are negative.

A TECHNICAL APPLICATION FOR THE HALL

The Hall effect can be used to establish plots of majority carrier concentration and mobility vs. temperature. This is an important tool for characterizing semiconductor devices. But the hall effect is also the basis for integrated circuit devices which measure magnetic fields. Hundreds of millions of integrated Hall circuits are in use, mainly as contactless switches, and mechanical proximity sensors Another application of a sensor is a current measuring transducer. The Hall effect integrated circuit allows the user to isolate the transducer from potentially high currents.

While providing current measurements within 1% accuracy.

Figure four is a schematic for a low voltage Ioff tester for a prototype semiconductor device. The device under test(DUT) is a high voltage semiconductor switching device. It is capable of conducting in excess of 350 Amperes with a forward voltage drop of less than 2 volts. In its non conduction state it is capable of blocking in excess of 2500 volts. Think of the DUT as a near lossless switch



The device has a great value to electrical systems which are designed to be as lossless as possible. A good example would be the space shuttle or future space station. A system would consist of many of these devices pulsing current from a battery source or generator and obtaining a usable voltage source. The system is complex and is controlled by a microprocessor. The switching speed and conduction time are crucial control processes and must be accurate. In order for the microprocessor to control the system, it must know how much current is passing **through** each device. Each device must have its own transducer which will relay current flow back to the microprocessor. There are two considerations when choosing the transducer:

1) The ideal transducer would be isolated from high current so that in the event of a device failure the microprocessor would not be damaged

2) Since the whole purpose of the system is to provide near lossless switching, the transducer must not consume a measurable amount of energy.

Using a resistor in series with the device violates condition 2. Even a .5 ohm resistor would dissipate 150 watts during conduction. Multiply that by the number of devices in the system which can be as great as 30 you have increased power wasted to 4500 watts during conduction. This would be unacceptable for applications pertaining to the space shuttle.

A Ilall sensor integrated circuit is optimal for this application. In terms of cost, lower end devices start at under three dollars, while upper end devices can exceed seventy. The power consumed by the Hall effect device is minimal. The voltage inputs vary from device to device, and range from 5 to 17 volts, Current drawn from the device generally do not exceed 100 milliamps.The power consumed by using this device is negligible compared to using a resistor. In addition the Hall effect sensor provides total isolation of microprocessor from the high current passing through the device.

The schematic is a resonant circuit. The DUT is controlled by a gate driver. When the device is turned on, current ramps up until the device is switched off by the gate driver. The current level is controlled by the pulse width of the gate signal. During conduction the forward voltage drop across the device is under 2 volts. When the DUT is switched off, the current drops to zero, and a resonant voltage signal develops across DUT anode to cathode. Depending on the current that was flowing through the device the resonant voltage may be as high as 2500 volts. The pulse rings down quickly, and the device waits for the next signal from the gate to conduct.

Construction of the circuit is now complete. We are waiting on delivery of several different Hall transducers to begin experimenting with.

The Hall effect device will be placed in proximity to the anode of the DUT. The current that travels through the printed circuit board trace (strip line) induces a magnetic field. It is imperative that the Hall effect transducer be placed such that the magnetic field of the strip line is perpendicular to the Hall device

There is an important consideration when using this device. Drift, or carrier mobility is a function of both time and temperature. The DUT is packaged as a module and the Hall effect sensor would be encapsulated in this package. The DUT generates a great deal of heat. Heat affects carrier mobility This phenomenon is a non linear response.

Once a Hall device is selected, extensive testing will have to be done to develop an understanding of how temperature affects drift. The Hall device output will be compared to Peason coil output. The Pearson coil is a length of copper wire wound around a torroid. It is an accurate way of measuring current Its use is practical in a laboratory environment. But due to cost considerations it is not a practical solution for measuring current in the module. Once data is collected and analyzed, a subroutine will have to be written for the microprocessor that will compensate for the drift phenomenon by introducing a scaling factor for the Hall voltage when temperatures are extreme.

We expect actual testing for carrier concentration and mobility vs. temperature to begin the second week of June. We will update this report once data has been collected and analyzed. Expected duration of testing is two weeks with at least one for characterization.